

CHINTAN P. ZAVERI

Deep Learning Engineer | Machine Learning Engineer

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Languages: English (Fluent) | German (Intermediate - B1)

SUMMARY

Highly motivated and experienced Machine Learning Engineer with a proven track record of success in developing and deploying cutting-edge AI solutions. Proven ability to design, train, and deploy machine learning models for a variety of applications, including object detection, classification, and 3D pose estimation. Expertise in a wide range of deep learning frameworks and technologies, including Python, PyTorch, TensorFlow, and OpenCV. Experience in creating production level systems using LLMs and TTS models, deployed at Genyad.com.

WORK EXPERIENCE

Smart Steel Technologies GMBH, Germany

Deep Learning Engineer | Feb 2023 – Present

- Researched and developed steel defect classification and detection models for real-time inference systems, resulting in a 15% increase in accuracy.
- Designed and implemented training and real-time inference pipelines for user onboarding, reducing onboarding time by 80%.
- Optimizing models for inference on edge devices and servers.
- Developed data preprocessing pipelines to optimize model performance, leading to a 10% improvement in efficiency.
- Collaborated with cross-functional teams to understand customer requirements and design software architecture for 3 different projects.
- Designed high-availability and performance architectures for AI systems, achieving 99.2% uptime.
- Architected an enterprise-wide document ranking and retrieval system by implementing the Model Context Protocol (MCP) to integrate Google Drive, GitLab, and Jira.
- Engineered complex multi-agent workflows using advanced orchestration and agent-to-agent communication, while optimizing model performance through targeted LLM fine-tuning.

Oxolo GMBH, Germany

Machine Learning Engineer | Nov 2020 – Jan 2023

- Developed and deployed AI-based talking avatars using voice puppetry, 3D face tracking, and audio-based facial animation, increasing user engagement by 25%.
- Designed and trained machine learning models for pose estimation, 3D pose retargeting, face tracking, and neural face rendering.
- Created training and inference pipelines for user onboarding and data preprocessing, enabling the onboarding of over 10,000 users.
- Deployed AI models on cloud platforms (AWS Sagemaker, Triton) for real-time and scalable applications, handling over 1000 requests per second.
- Created models for photo editing: Background removal, super resolution, and trained diffusion-based models for product image generation.

University of Bonn, Germany

Research Assistant | Oct 2019 – Oct 2020

- Conducted research on human motion forecasting in the Computer Vision group, contributing to the publication of a research paper.
- Developed and evaluated pose forecasting models using deep learning techniques, achieving a 5% improvement over existing methods.
- Processed and analyzed large datasets of human motion data.

DHL, Bonn, Germany

Data Scientist Intern | Feb 2019 – Aug 2019

- Performed data analytics for onboarding 8 different countries with different languages, resulting in a 10% increase in successful onboarding.
- Tested and improved features in deployed systems, identifying and resolving critical bugs.
- Conducted explorative and systematic analyses of geocode data for over 50 countries.
- Developed and tested approaches for normalizing addresses, improving address accuracy by 20%.

Genesis Artificial Intelligence P. L., India

Machine Learning Engineer | May 2018 – Aug 2018

- Developed and deployed a chatbot for clients using natural language processing and machine learning techniques, achieving a 90% user satisfaction rate.
- Validated various NLP and ML modules for chatbot development, ensuring high accuracy and performance.

SKILLS

Programming Languages: Python, C++, C, Java, JavaScript, PHP, Node.js, AngularJS, MeteorJS, jQuery

Deep Learning Frameworks: PyTorch, TensorFlow

Computer Vision Libraries: OpenCV

Databases: MySQL, NoSQL, MongoDB, Clickhouse

Other: Git, CI/CD, Docker, AWS, CUDA, OpenGL, Data Structures and Algorithms, Image Processing, Natural Language Processing

EDUCATION

Masters in Computer Science – Focus on Computer Vision and Graphics

University of Bonn, Germany | 2018-2020

Bachelors in Computer Science

The Maharaja Sayajirao University of Baroda, India | 2014-2018

PROJECTS

Genyad: Video ad variations using LLMs – Implemented RAG-based video ad composer from old videos. Using CV to split videos and create new engaging video ads using LLMs.

Human Motion Forecasting using GANs with PyTorch

Robo Sorting – Sorting of objects using robotic arm with C++ and OpenCV

Border Surveillance System – Capturing and processing video from multiple cameras for movement detection

Image Segmentation – Using watershed and snakes algorithm

PUBLICATIONS & AWARDS

"Intention-based Long-Term Human Motion Anticipation" (under supervision of Prof. Dr. Juergen Gall) published in the International Conference on 3D Vision 2021

"WBC segmentation using watershed and snake algorithms for diagnosis of Malaria" published in IJSRD Nov'17 issue.

OPEN-SOURCE PROJECTS

PhpMyAdmin: Contributed to the development of PhpMyAdmin, gaining experience in working with a global team and using GitHub.

BRL-CAD: Learned the basic BRL-CAD engine (CSG-based) and gained hands-on experience with Three.js (WebGL) library.

EXTRACURRICULAR

Brown Belt in Shotokan Karate and basic training in Taekwondo. Participated in various national and state-level Kumite and Kata competitions.

Trained Tabla player (Musical Instrument).